

L-2-3: Useful Statistical Procedures for Nowcasting, Part 3: Seasonal Adjustment

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Seasonal Adjustment of Time-Series: Introduction

Economic variables are often influenced by systematic and recurrent (within-a-year) patterns owing to weather and social factors -- more commonly referred to as 'seasonal effects.'

Seasonal adjustment involves the removal of the seasonal effects from a time-series.

Why bother? Because seasonal adjustment allows for a more informed, logical assessment of current economic conditions.

Seasonal Adjustment: Introduction

Seasonal adjustment makes it easier to identify non-seasonal developments such as long-term secular movements (i.e., trend), cyclical variations, or irregular factors, which carry the most important economic signals.

In practice, these components are unobserved and must be identified from the raw data based on *a priori* assumptions.

Note that seasonal adjustment does not remove the impact of irregular events from a time-series.

It follows that if the irregular components of a series are large, a seasonally adjusted series may not be “smooth”.

Seasonal Adjustment: Learning Objectives

The aim of this module is to review the main principles of seasonal adjustment, and the fundamental steps of the X13 seasonal adjustment procedure as implemented by the U.S. Census Bureau.

We will also learn how to invoke this routine from EViews.

Main Assumptions of Seasonal Adjustment

For seasonal adjustment purposes, it is generally assumed that a time-series is made up of four unobserved components:

a) A trend-cycle component (TC);

b) A seasonal component (S);

c) A calendar component (C), and

d) An irregular component (I).

Each component needs to be identified (and estimated) from a time-series using so-called "signal extraction" techniques.

The Additive versus the Multiplicative Model

The time-series model for seasonal adjustment may be either

$$Additive: \ TS_t = \ TC_t + \ S_t + \ C_t + \ I_t$$

or

$$Multiplicative: \ TS_t = \ TC_t x \ S_t x \ C_t x \ I_t$$

Trend-Cycle Component

The trend-cycle component (T_t) is the underlying path of the series. It includes both the long-term trend and the business-cycle movements in the data.

The long-term trend is typically associated with structural changes in the economy, such as population growth and changes in technology and productivity.

Business cycle variations are related to the periodic oscillations of different phases of the economy (i.e., recession, recovery, growth and decline) that generally repeat themselves with a period of between two and eight years.

Seasonal Component

The seasonal component (S_t) includes those fluctuations that repeat themselves with similar annual timing, direction and magnitude.

The seasonal component relates to climate factors, administrative or legal rules, and social/cultural traditions and conventions – including calendar effects that are stable in annual timing (e.g., public holidays or other national festivities).

Calendar Component

The calendar component (C_t) comprises effects that are related to the different characteristics of the calendar from period to period.

Calendar effects are both seasonal and non-seasonal, and only the *non-seasonal* part should be included in the calendar component.

The most common calendar effects are:

a) Trading-Day or Working-Day Effect.

b) Moving Holiday Effect

c) Leap-Year Effect

Trading-Day Effects

The trading-day effect detects the different number of specific days within a quarter (M-T-W-T-F-S-S) relative to the standard weekday composition of a quarter.

The working day-effect captures the differences between the number of working days (e.g., Monday to Friday) and the number of weekend days (e.g., Saturday and Sunday) in a quarter.

The trading-day effect assumes an underlying pattern associated with each day of the week.

The working-day effect postulates different behavior between groups of weekdays and weekends.

Both effects need to account for national holidays.

Moving Holiday and Leap-Year Effect

A moving holiday is associated with events of religious or cultural significance withing a country that change date from year to year (e.g., Easter or Ramadan)

The leap-year effect is needed to account for the extra day in February of a leap year, which may generate a four-year cycle with a peak in the first quarter of each leap year.

Irregular Component

The irregular component (I_t) captures all the other fluctuations that are not part of the trend-cycle, seasonal, and calendar components (i.e., the unexplained residuals). These effects are characterized by the fact that their timing, impact, and duration are unpredictable.

The irregular component includes the following effects:

a) Outlier effects representing abrupt changes in the series, sometimes related to unexpected weather or socioeconomic effects (such as natural disasters, strikes, or economic or financial crises).

b) White noise effects.

In the absence of outliers, the irregular component is assumed to be N(0,1) random variate.

Outlier Effects

The three main types of outliers are often used for economic time-series:

a) Additive outlier: an outlier that relates to a single period

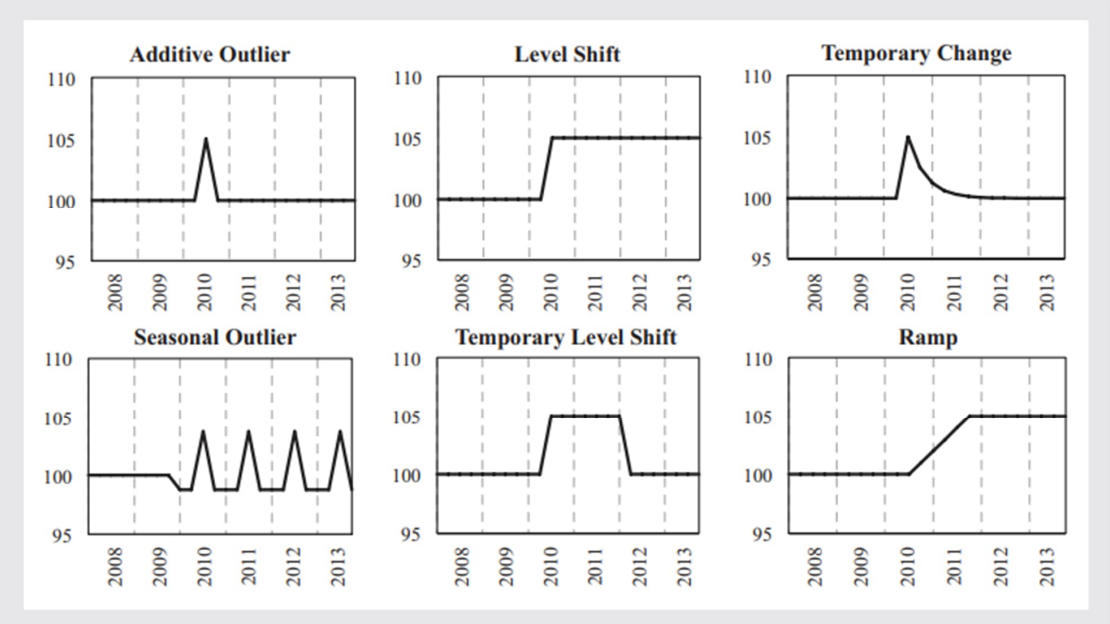
b) Level shift: an outlier that changes the level of a series permanently

c) Transitory change: the effects on the series fade out over several periods.

Other effects are seasonal outliers (which affect only certain quarters/months of the year), ramp outliers (which allow for a linear increase or decrease in the level of a series), or temporary level shifts.

The different types of outliers are shown on the next slide.

Types of Outliers



Two Important Caveats

First, seasonal adjustment is not meant for smoothing series.

A seasonally adjusted series is the sum of the trend-cycle component and the irregular component. When the irregular component is strong, the seasonally adjusted series may not present a smooth pattern over time.

Second, seasonally adjusted series complement the original data, but can never replace the original data for the following reasons:

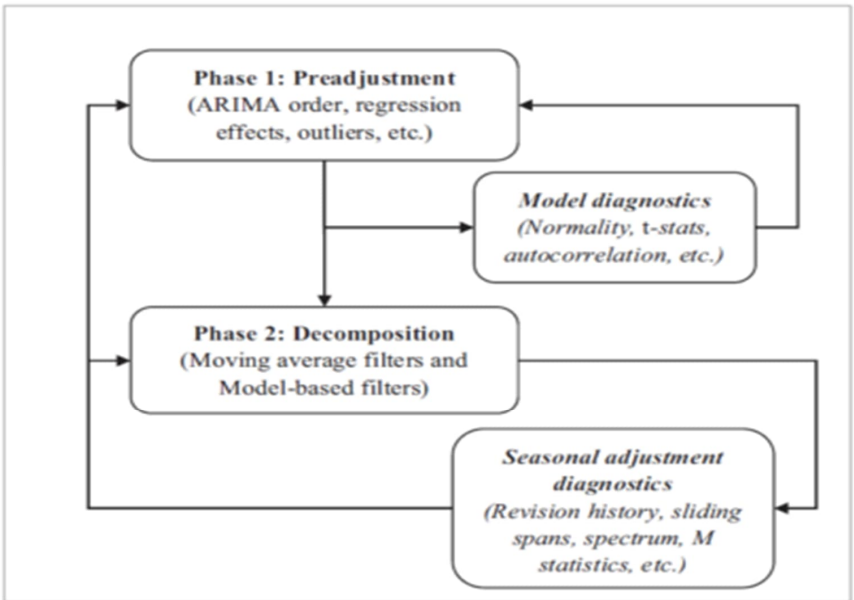
a) Unadjusted data are useful. Calculation of seasonally adjusted data, by definition, involves a loss of information.

b) No unique solution exists on how to conduct seasonal adjustment

c) Seasonally adjusted data are subject to revision even if the actual data has not changed.

Main Elements of Seasonal Adjustment Procedure

Box 7.2 Main Elements of Seasonal Adjustment Procedures



[1] Diagram based on Findley and others (1998).

Seasonal Adjustment Procedure: Pre-Adjustment (Phase 1)

Modern seasonal adjustment procedures follow a two-stage procedure.

The first stage or phase is called “pre-adjustment”, where the aim is to select a regression model with ARIMA errors that best describes the characteristics of the original series.

The selected model is used to adjust the series for deterministic effects and to extend the series backwards (“back-casts”) and forwards (“forecasts”) to be used in the second stage (the time-series decomposition process).

The pre-adjustment phase comprises mainly the choice of how the individual components are related to each other (additive or multiplicative), the order of the ARIMA model, calendar effects, and outliers and other intervention variables.

Seasonal Adjustment Procedure: Decomposition (Phase 2)

The second stage or phase of the seasonal adjustment process involves *decomposing* the series into the unobserved components.

The series adjusted for deterministic effects is decomposed into three unobserved components: trend, cycle and irregular.

Although the two steps are treated separately, they should be considered fully integrated in any seasonal adjustment procedure. Different choices in the pre-adjustment phase yield different decomposition results.

Also, the results of the decomposition phase may ultimately necessitate modifications to the model in the pre-adjustment phase.

The results of both stages need to be carefully assessed to determine whether the seasonal adjustment results are acceptable.

Additive Model

The initial step in the pre-adjustment phase is to decide on the type of model that will be used to seasonally adjust the series in the decomposition phase of the procedure (i.e., stage 2).

The choice is usually between the *additive* model and the *multiplicative* model.

For the additive model, the original series $X_t = \mathrm{T_t} + \mathrm{S_t} + \mathrm{C_t} + \mathrm{I_t}$ and the unobserved components are assumed to be mutually independent from one another.

Moreover, the seasonally adjusted data is $X_t^a = X_t - (S_t + C_t)$

The additive approach is suited for series where the seasonal pattern appears to be stable over time and does not evolve in accordance with the movements in the trend.

Multiplicative Model

In the multiplicative model, the series is decomposed as the product of the unobserved components: $X_t = \mathrm{T_t}\ x\ \mathrm{S_t}\ x\ \mathrm{C_t}\ x\ \mathrm{I_t}$. Thus, the magnitude of the individual components is proportional to level of the series.

Because the trend-cycle component determines the overall level of the series, the other unobserved components are expressed as percentages of T_t.

The seasonal and calendar adjusted series for the multiplicative model is

$$X_t^a = \frac{\mathrm{X_t}}{\mathrm{S_t} \cdot C_t} = T_t \cdot I_t$$

Pre-Adjustment (Phase 1): ARIMA Model

Once the choice between additive and multiplicative is made, the next step is to identify an ARIMA model, with or without deterministic variables present (often called the "regARIMA" model).

The ARIMA model for seasonal time-series can be written as:

$$\emptyset(B)\Phi(B^S)(1-B)^d(1-B^s)^D X_t = \theta(B)\Theta(B^S)\varepsilon_t$$

where

X_t is the original series

B is the lag operator, which is defined by $X_{t-1} = BX_t$

s is the seasonal frequency, 4 for quarterly series; 12 for monthly series

Pre-Adjustment (Phase 1): ARIMA Model

$\emptyset(B) = 1 - \emptyset_1 B - \cdots - \emptyset_p B^p$ is the regular autoregressive (AR) operator of order p;

$\Phi(B^S) = 1 - \Phi_1 B^s - \cdots - \Phi_P B^{sP}$ is the seasonal AR operator of order P;

$\theta(B) = 1 - \theta_1 B - \cdots - \theta_p B^q$ is the regular moving average (AR) operator of order q;

$\Theta(B^S) = 1 - \Theta_1 B^s - \cdots - \Theta_P B^{sQ}$ is the seasonal MA operator of order Q; and,

ε_t is a white-noise process

Pre-Adjustment (Phase 1): ARIMA Model

Identifying an ARIMA model consists of estimating the orders of the AR parameters, the MA operators, and well as establishing the non-seasonal and seasonal integration orders.

A seasonal model for quarterly series, for example, is usually indicated as $(p, d, q)(P, D, Q)_4$.

Note that "d" is the differencing order for the actual series, viz. $(1 - B)^d$, while D is the differencing order for the seasonal difference, namely $(1 - B^s)^D$.

Typical values for d and D are unity.

Pre-Adjustment (Phase 1): Calendar effects

Calendar effects need to be removed from the series because their presence could affect negatively the quality of decomposition into unobserved components.

For example, when a month contains more working days that 'normal', macroeconomic series economic activities may present a spike in that month, when all that happened is that there were more days available for production.

To avoid such distortions, calendar effects should be removed from the series before decomposition takes place.

This is done using special deterministic regressors (independent, exogenous) in the ARIMA regression model. The most frequently used calendar regressors are:

Pre-Adjustment (Phase 1): Trading Day Effects

Trading day effects, which are are captured by the following six regressors:

1. $td_t^1 = (\#Mondays - \#Sundays)$
2. $td_t^2 = (\#Tuesday - \#Sundays)$
3. $td_t^3 = (\#Wednesday - \#Sundays)$
4. $td_t^4 = (\#Thursday - \#Sundays)$
5. $td_t^5 = (\#Friday - \#Sundays)$
6. $td_t^6 = (\#Saturdays - \#Sundays)$

The assumption is that each day of the week may influence the underlying process with different magnitudes – that is, their underlying regression coefficients are different.

Pre-Adjustment (Phase 1): Working Day Effects

The working day effect is measured by a single regressor that compares the group of workdays (e.g., Monday to Friday) with the group of weekend days (e.g., Saturday and Sunday) using the following equation:

$$wd_t = \left(\#Weekdays \; - \left(\frac{5}{2}\right) \#Weekend\ Days \right)$$

The 5/2 factor is required to make the working-days regressor zero over a regular seven-day week composition.

When $\boldsymbol{wd_t}$ is greater than 0 it means that the month/quarter t has more working days than a standard month/quarter.

Again, this approach assume that all the weekdays have similar effects (in sign and value) but are different from weekend day effects.

Pre-Adjustment (Phase 1): Easter Effects

The Easter date moves between March (Q1) and April (Q2).

The Easter regressor assumes that the level of activity changes on the wth day before the Easter holiday (for a given w) and remains at the new level until the day before Easter.

The Easter regressor calculates the proportion of the w days falling in March (Q1) and April (Q2) respectively.

Pre-Adjustment (Phase 1): Easter Effects

After defining the length of the Easter effect, i.e., setting a value for w, the regressor is calculated as follows:

$$e_t = \frac{W_t}{w} - \bar{W}$$

where

W_t = number of w days before Easter falling in month/quarter t (e.g., in March)

$\bar{W}$ = the long-term proportion of days in month/quarter t (usually set to 0.5 for both March (Q1) and April (Q2))

For example, suppose the number of days is assumed to be 15 (= w). If 7 of those days fall in March, then W_t = 7 and the $e_t = \frac{7}{15} - 0.5$.

Pre-Adjustment (Phase 1): Leap Year Effects

The leap year effect is captured by

$$ly_t = \begin{cases} 0.75, & \text{if t is February of leap year} \\ -0.25, & \text{if t is February of a non} - \text{leap year} \\ 0, & \text{otherwise.} \end{cases}$$

The regressor ly_t reproduces the deterministic four-year cycle with a peak in February of the leap years.

Note that over a 4-year period, the leap year effect is fully compensated by the negative effects in the subsequent non-leap years.

Pre-Adjustment (Phase 1): Calendar Effects

Calendar effects should be performed only for those series for which there is both statistical and economic justification/interpretation of the calendar effects.

Specifically, the sign of the regression coefficient should be interpretable from an economic standpoint (e.g., the leap year effect should always be positive, the Easter effect should be positive for consumption of tourism-related services and negative for production-based series).

Pre-Adjustment (Phase 1): Calendar Effects; Final Remarks

If the estimated coefficient of the calendar effect is statistically insignificant or of the wrong sign, you should not adjust the series for such calendar effects.

Calendar effects are more likely to be noticeable in monthly series rather than quarterly.

Lastly, the standard calendar effect variables defined above may not be appropriate for all countries.

You will likely need to define your own deterministic variables to capture the calendar effects in your country and include them in the regression as "user-defined" variables (see below).

Pre-Adjustment (Phase 1): Outliers and Intervention Variables

Outliers can lead to significant distortion in the decomposition of time-series variables.

For example, unexpected extreme weather conditions can seriously affect output of agricultural crops. The sudden fall in agricultural activity should be allocated to the irregular component of the series, without affecting the long-term trend its seasonality factors.

To achieve this, the seasonal adjustment routine will normally remove outliers from the original series before the seasonal adjustment process begins. The outliers will be re-introduced after the seasonal adjustment process has been completed.

Most seasonal adjustment routines have a procedure for the automatic detection of additive outliers, temporary change outliers, and level shifts.

Time-Series Decomposition Methods: X-11 Filter/SEATS Filter (Phase 2)

The X-11 and SEATS filters are two popular methods for identifying the 4 key components of a *pre-adjusted* time-series.

Both X-11 and SEATS apply symmetric filters to the pre-adjusted series to derive estimates of trend-cycle, seasonal, and irregular components.

The formulas used, however, differ significantly from one another. We shall focus on the X-11 filter in what follows. Discussing the exact nature of these formulas is beyond the scope of this lecture (see the references for precise details). Kindly refer to the documentation for a detailed explanation of the complementary SEATS filter.

X-11 Filter (Phase 2)

The X-11 method is an iterative process, using appropriate moving averages to decompose a time series into trend/cycle, seasonal, and irregular components:

The basic algorithm of the X-11 method will be presented for a *monthly* time series X_t that is assumed to be decomposable into trend, seasonality and irregular component according to an *additive* model $X_t = T_t + S_t + I_t$.

Notation: MA_{nxm} Moving Average

An MA_{nxm} moving average is a m-term moving average simple moving average taken over n consecutive sequential spans:

$$\bar{X}_t^{3x4} = \frac{1}{3}\left(\frac{X_{t-3} + X_{t-2} + X_{t-1} + X_t}{4}\right)$$

$$+ \frac{1}{3}\left(\frac{X_{t-2}+X_{t-1}+X_t+X_{t+1}}{4}\right)$$

$$+ \frac{1}{3}\left(\frac{X_{t-1}+X_t+X_{t+1}+X_{t+2}}{4}\right)$$

Note: the coefficients on each unique series are $\frac{1}{12}$ (1,2,3,3,2,1)

X-11 Algorithm: Steps 1 and 2 (First Iteration)

Step 1: Estimation of Trend/Cycle component by a 2 × 12 *moving average:*

$$TC_t^1 = MA_{2x12}(\varepsilon_t)$$

where ε_t is the *pre-adjusted* series (as generated in Stage 1).

Step 2: Estimation of the Seasonal-Irregular component:

$$(S_t + I_t)^1 = \varepsilon_t - TC_t^1$$

X-11 Algorithm: Step 3 (First Iteration)

Step 3: Estimation of the Seasonal S_t^1 component by 3 x 3 moving average:

$$S_t^1 = M_{3x3}\left[\left(S_t + I_t\right)^1\right]$$

The moving average used here is a 3×3 moving average over 5 terms, with coefficients $\frac{1}{9}\{1,2,3,2,1\}$.

The seasonal component is then centered (normalized) using a 2×12 moving average:

$$\tilde{S}_t^1 = S_t^1 - M_{2x12}(S_t^1)$$

X-11 Algorithm: Step 4 (First Iteration)

Step 4: Estimation of the seasonally adjusted series:

$$SA_t^1 = (TC_t + I_t)^1 = \varepsilon_t - \tilde{S}_t^1$$

This first estimation of the seasonally adjusted series must, by construction, contain less seasonality.

The X-11 method again executes the algorithm presented above (i.e., iterates), changing the moving averages to allow for the reduced level of seasonality in the second (and subsequent) iteration.

X-11 Algorithm: Step 5 (Second Iteration)

Step 5: Estimation of Trend by 13-term "Henderson" moving average:

$$TC_t^2 = H_{13}(SA_t^1)$$

Henderson moving averages are trend filters, commonly used in time series analysis to smooth seasonally adjusted estimates in order to generate a trend estimate.

While they do not have special properties in terms of eliminating seasonality (limited or none at this stage), they are known to have very good smoothing power.

X-11 Algorithm: Steps 6 and 7 (Second Iteration)

Step 6: Estimation of the Seasonal – Irregular Component:

$$(S_t + I_t)^2 = \varepsilon_t - TC_t^2$$

Step 7: Estimation of the Seasonal component by 3×5 *moving average over each month:*

$$S_t^2 = M_{3x5}[(S_t + I_t)^2]$$

The moving average used here is a 3×5 moving average over 7 terms, of coefficients {1/15, 2/15, 3/15, 3/15, 3/15, 2/15, 1/15} and retains linear trends.

The coefficients are then normalized such that their sum over the whole 12-month period is approximately cancelled out:

$$\tilde{S}_t^2 = S_t^2 - M_{2x12}(S_t^2)$$

X-11 Algorithm: Step 8 (Second Iteration)

Step 8: Estimation of the seasonally adjusted series:

$$SA_t^2 = (TC_t^2 + I_t)^2 = \varepsilon_t - \tilde{S}_t^2$$

Note that the main difficulty of the X-11 routine lies in the choice of the moving averages used for the estimation of the trend in steps 1 and 5, and in the estimation of the seasonal component in steps 3 and 5.

X11 Filter: Assessment

In terms of actual usage, the key issue is the quality of the results derived from the X-11 filter.

A quick way to assess the X-11 results is to examine the final seasonal adjustment ratios to the seasonal adjustment ratios with the irregular factors.

Seasonality is expected to be stable over time.

X-11 Filter: Assessment

If the SI ratios are too volatile relative to the seasonal factors, this indicates that the series contains a strong irregular component and that the seasonal effects have absorbed too much volatility.

This implies the need to use shorter filters to extract the seasonal effects from the seasonal-irregular ratios.

That said, longer filters are preferred for extracting stable seasonal factors.

Seasonal Adjustment Using EViews

EViews offers an easy-to-use front-end for working with the U.S. Census Bureau's X-13 seasonal adjustment tools.

In addition to providing a wide range of new features, X-13 can perform updated versions of X-11/X-12 or TRAMO/SEATS ARIMA seasonal adjustment methods.

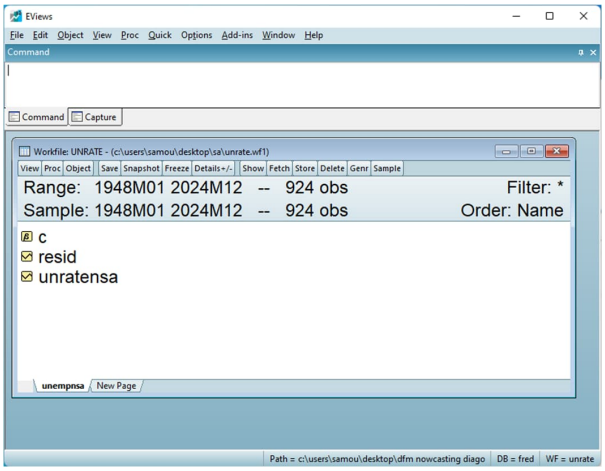
When you perform X-13 seasonal adjustment, EViews carries out the following steps:

- writes out an X-13 specification file and data files for the series
- executes the X-13 program in the background, using the contents of the specification file
- reads back the output file and processed data into your current EViews workfile

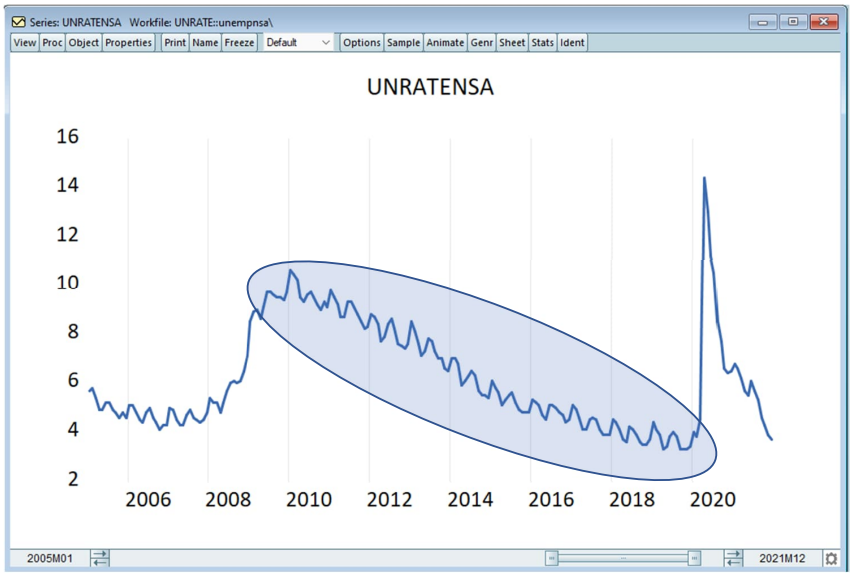
Invoking the X13 routine in EViews

The first step is to open a spreadsheet view of the series you wish to seasonally adjust. You can double-click on the series name or use the show command followed by the name of the series in the command line area.

Consider the following workfile, which can be downloaded from the Moodle course site. It contains a single series: the non-seasonally adjusted unemployment rate for the USA, *unratensa*, which is a monthly series spanning 1948M1-2021M12.



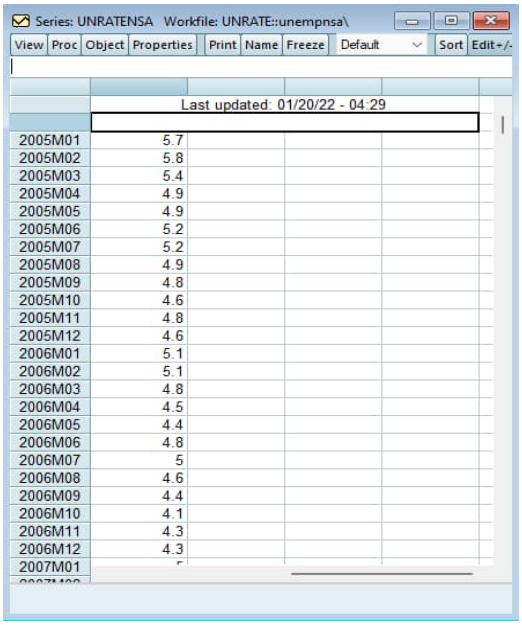
NSA Unemployment Rate, 2005M1-2021M12



The non-seasonality of the monthly unemployment series is quite evident.

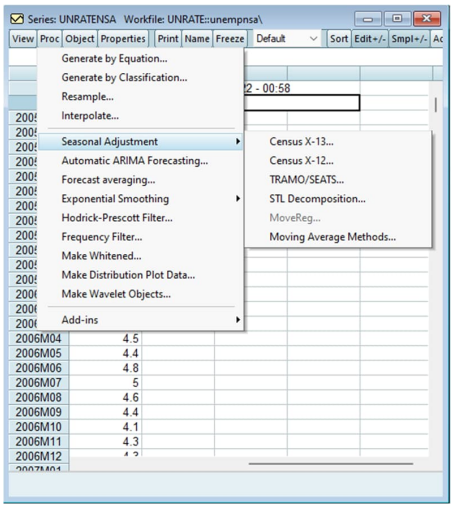
NSA Unemployment Rate, 2005M1-2006M12

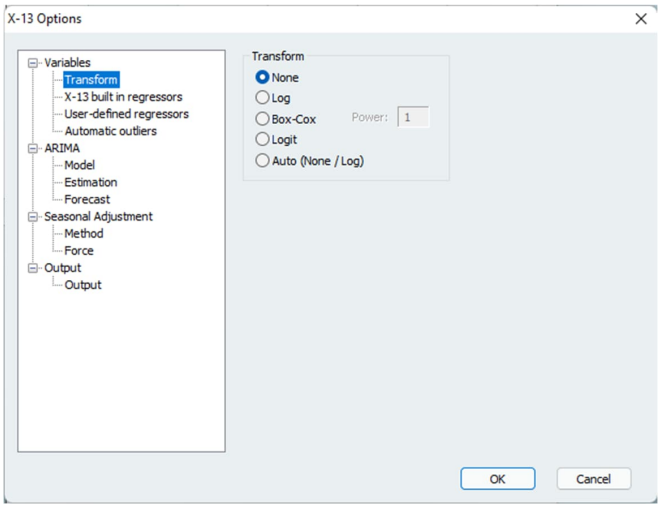
Double-click on the *unratensa* series to obtain a spreadsheet view of the data, showed below starting with the 2005M1 observation.



Invoking the X-13 Routine in EViews

The X-13 routine can be invoked from series proc menu: "Proc→Seasonal Adjustment→Census X-13…", which brings up the X-13 Options dialog box (on the right).





The RHS options allow you to control how the X-13 seasonal adjustment is performed. We will discuss each option briefly in what follows.

Transform Option

The Transform settings allow you to transform the source series and regressors *before* fitting an ARIMA model. When you click on the transform entry you will be prompted to choose a transformation method.

You may choose between no transformation (None), or a natural log (Log), Box-Cox (Box-Cox), logistic (Logit), or automatically selected (Auto) transformation.

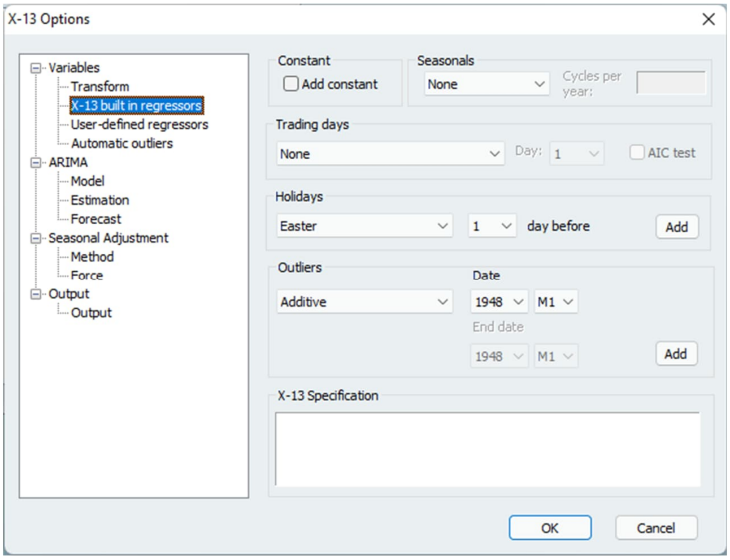
The Auto option (the recommended setting) chooses between no transformation and a log transformation based on the Akaike information criterion.

The logit option transforms the series y_t to $\log\left(\frac{y_t}{1-y_t}\right)$. It is defined for series with values that are strictly between 0 and 1. For the Box-Cox option, you must provide the parameter value λ for the transformation:

$$\begin{cases} \log(y_t) & if\ \lambda = 0 \\ \lambda^2 + \dfrac{y^\lambda - 1}{\lambda} & if\ \lambda \neq 0 \end{cases}$$

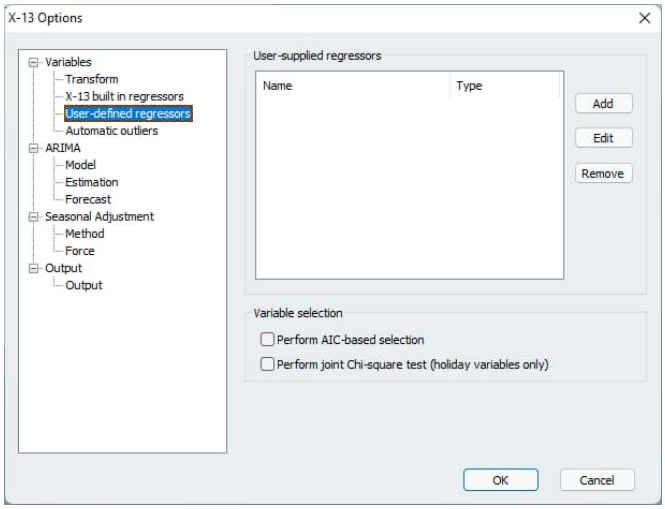
X-13 Built-In Regressors

The X-13 built in regressors page lets you specify (and add) any of the X-13 built-in regressors for the ARIMA regression model, including trading day effects, holidays, and outliers.



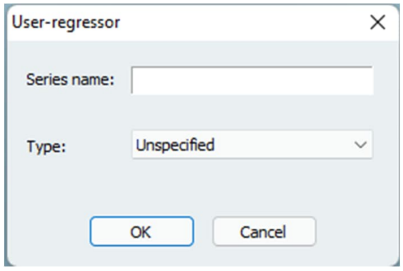
X-13 User-defined regressors

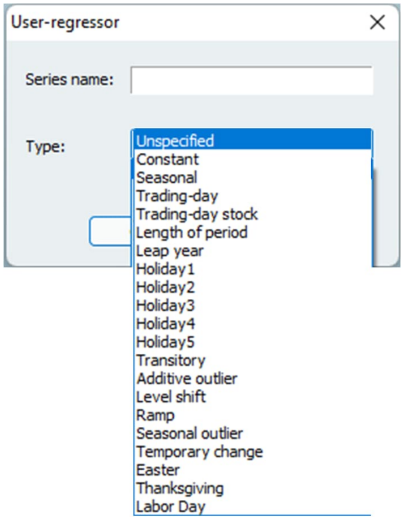
The User-defined regressors page allows non-X-13 built-in variables to be used as regressors in the ARIMA model. This is especially useful for non-USA applications.



Clicking on the **Add** button brings up a dialog asking you to type in the name of the workfile (user-defined) variable that you would like to include as a regressor.

X-13 User-defined regressors

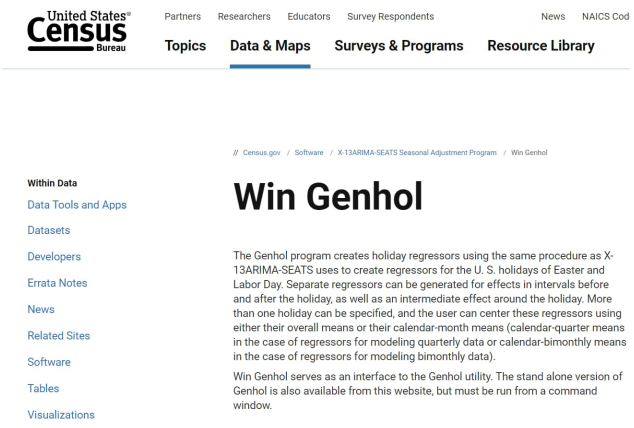




You may enter any valid EViews series name or expression, such as “X” or “log(X)”, or “@pch(X)” in the Series name edit field.

You will then be asked to specify the type of the user defined regressor being added to the ARIMA regression.

Creating Holiday Regressors -- the Genhol Program



Website: [Win Genhol (census.gov)]

- The Genhol program, from the United States Census Bureau, creates holiday regressors using the same procedure as X-13ARIMA-SEATS uses to create regressors for the U.S. holiday of Easter and Labor Day.
- Separate regressors can be generated for effects in intervals before and after the holiday, as well as an intermediate effect around the holiday.

Creating Holiday Regressors -- the Genhol Program

This example will create a file containing user-defined regressors for Chinese New Year and Idul Fitri (an Islamic holiday that marks the end of Ramadan)

```
global{
    numhol = 2
    outfile = "indhol.dat"
    outspec = "indhol.reg"
    usergroup = yes
    chi2test = yes

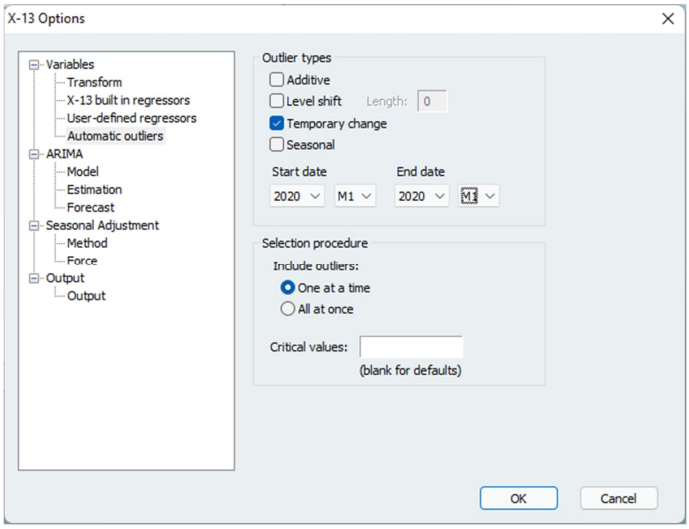
}
holiday1{
    name = CNY
    begbefore = -10
    endbefore = -1
    begafter = 5
    endafter = 14
    infile = "cnyInd.dat"
    center = calendar
}
holiday2{
    name = IdulFitri
    begbefore = -42
    endbefore = -21
    begafter = 1
    endafter = 7
    infile = "IFHolInd.dat"
    center = mean
}
```

```
regression{
   user=(
    BeforeCNY    BetweenCNY    AfterCNY
    BeforeIdulFitri    BetweenIdulFitri
AfterIdulFitri
   )
   file="indhol.dat"
   format="datevalue"
   start=1970.1
   usertype=(
    holiday     holiday     holiday
    holiday2    holiday2    holiday2
   )
   chi2test=yes
}
```

This will generate six holiday regressors – three Chinese New Year regressors, and three Idul Fitri (for before, after and intermediate holiday effects).

Automatic Outliers

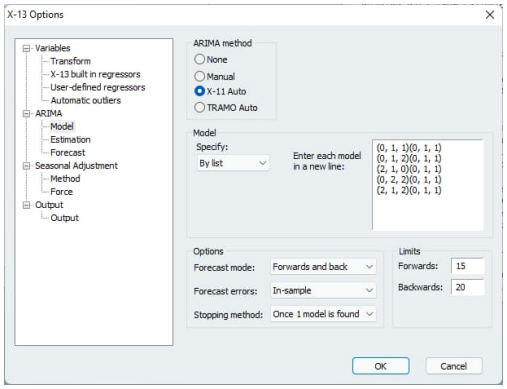
The “Automatic outliers” entry allows specification of automatic outlier detection routines in the X-13 package. Note that the default is not to allow for any automatic detection of outliers.



The above settings test for outlier in the unemployment rate during 2020 – i.e., during the COVID-19 pandemic.

X-13: ARIMA Section

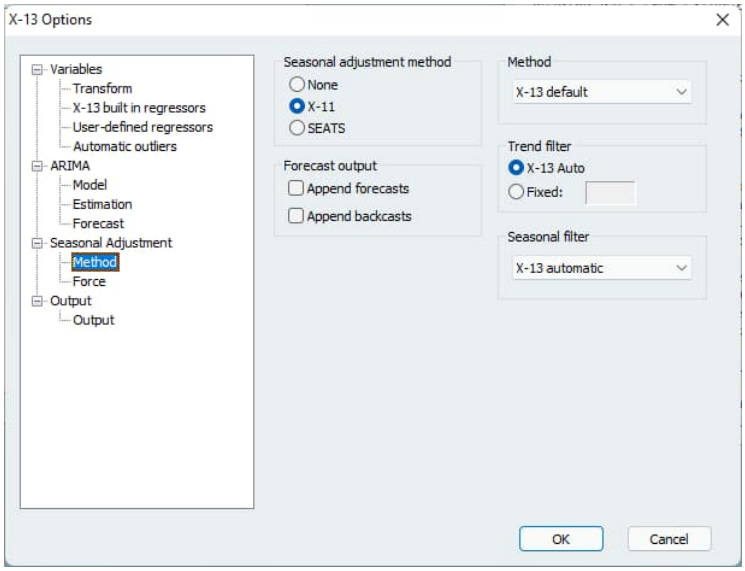
The ARIMA branch controls whether an ARIMA model is fitted to the data in the pre-adjustment phase. The X-11 Auto option (i.e., where the program attempts to find the best ARIMA model to use) is recommended for an initial run, and with forecasts and back casting to support the X-11 phase.



The Estimation branch (not shown) allows one to control the estimation options for the ARIMA model, while the Forecast branch controls the amount of forecasts/back casts produced by the ARIMA model once it is estimated.

X-13: Seasonal Adjustment Options

The Seasonal Adjustment branch of the dialog contains the options for the actual seasonal adjustment process undertaken by X-13.



X-13: Seasonal Adjustment Options

You may use the radio buttons in the Seasonal adjustment method section to select either X-11 or SEATS based seasonal adjustment. X-11 is the default.

Alternatively, you may choose None to perform no seasonal adjustment so that X-13 simply estimates the specified ARIMA regression model and optionally produces forecasts.

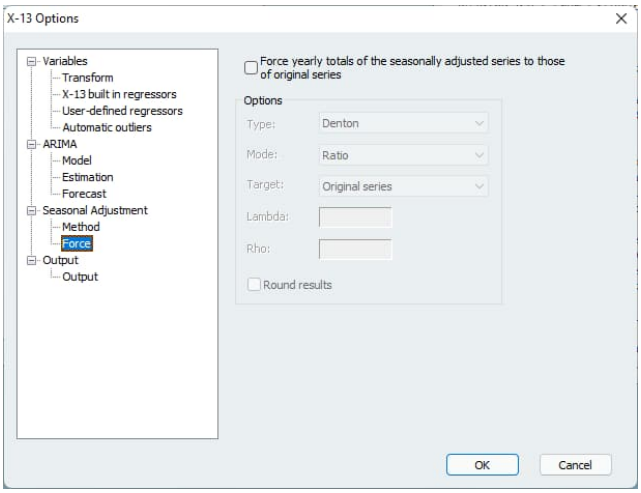
If you specify a seasonal adjustment method, EViews will display a set of Forecast output options.

The two checkboxes allow you to instruct X-13 to append one year of forecasted and/or backcasted values to certain output series (e.g., the seasonal factors D10, if performing X-11 adjustment and the seasonal factors S10, if performing SEATS adjustment), and to display the results in the table output.

X-13: Force Options

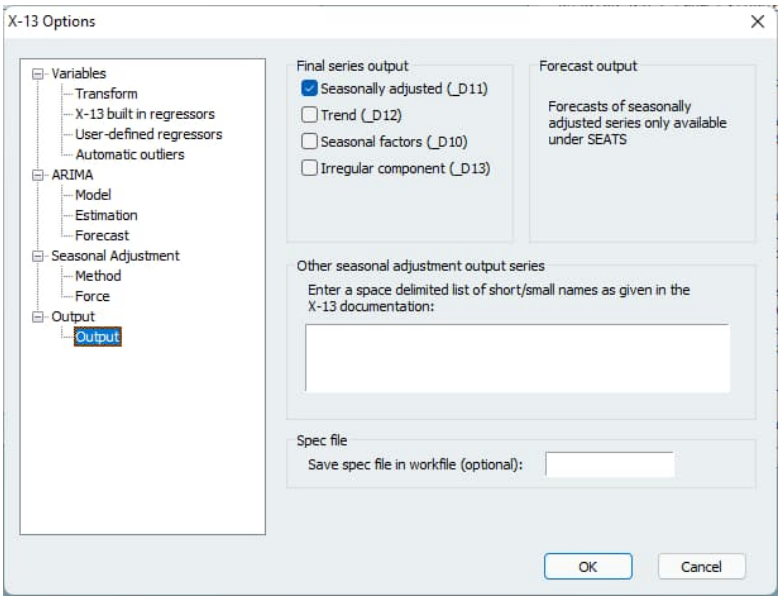
The Force section provides the ability to force annual totals of the adjusted series to match those of the original series.

X-13 provides two methods for forcing the totals; the original Denton method of X-11, and a newer regression benchmarking method of Cholette and Dagum (1994).



X-13: Output Options

The Output section allows you to set which output series will be saved to the workfile.



X13: Output Options

When performing either X-11 or SEATS seasonal adjustment, you may elect to output the seasonally adjusted data (called D11 in X-11 short name terminology, and S11 in SEATS), the trend component (D12, S12), seasonal factors (D10, S10) or the irregular component (D13, S13) by selecting the corresponding check boxes in the Final series output section.

Each of these output series will be imported into EViews with a name given by the original series name plus an underscore followed by the X-13 short name.

For example, if your original series is called GDP, the X-11 seasonal factors will be added to the workfile with the name GDP_D10.

Lastly, you may optionally enter a name in the Spec file box to save the X-13 specification file as a text object in your workfile. Saving the specification file lets you reuse or modify the specification for further use (note that you can specify X-13 via the command line using a text object in the workfile).

X13 Example: U.S. Unemployment Rate

As an example of using the EViews X-13 proc, let's seasonally adjust monthly unemployment data in the "unrate.wf1" workfile.

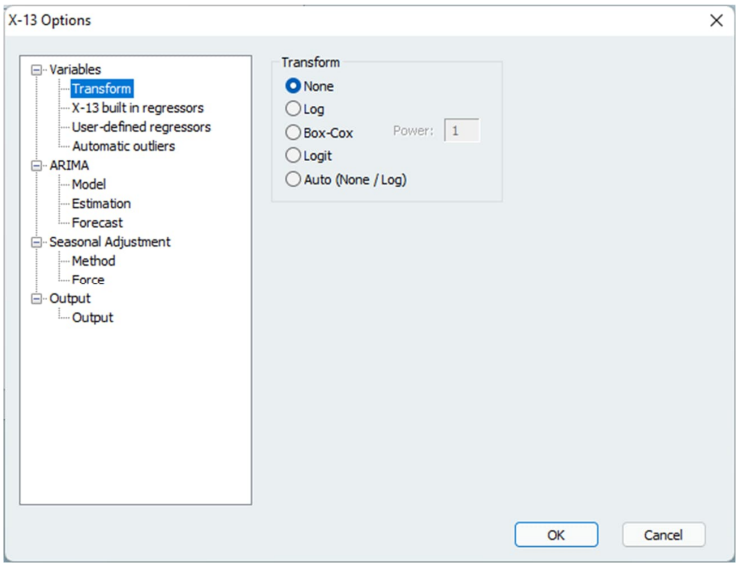
It contains monthly non-seasonally adjusted US unemployment data from January 1948 to December 2021 in a series called UNRATENSA.

To perform X-11 based seasonal adjustment with automatic ARIMA selection using X-11-Auto, open the UNRATENSA series and click on Proc/Seasonal Adjustment/Census X-13.

EViews will display the X-13 dialog…

X13 Example: U.S. Unemployment Rate

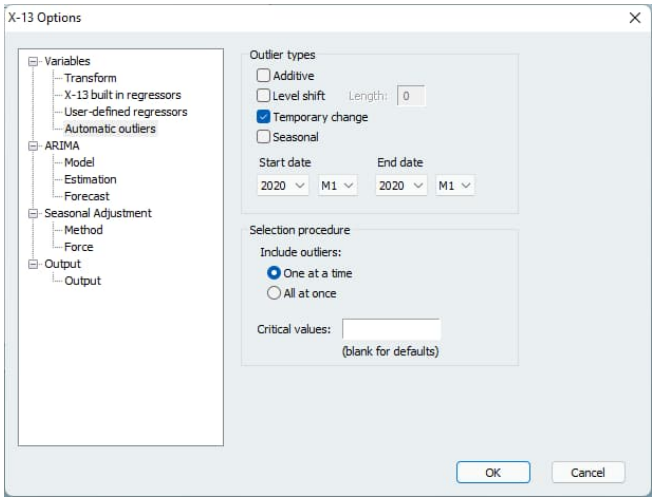
Transform Setting:



Note that this example will not be using any exogenous variables or transformations so we can jump to the Automatic Outlier section.

X13 Example: U.S. Unemployment Rate

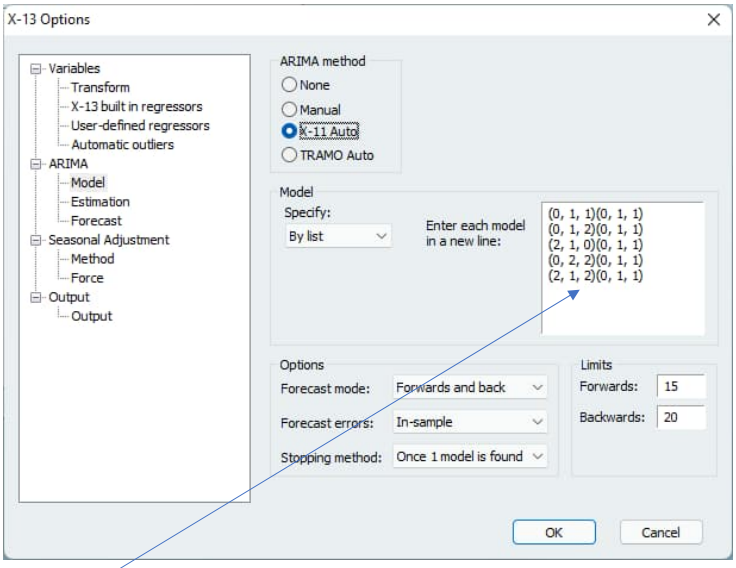
Automatic Outliers Setting:



We allow for the possibility of an outlier during the COVID period, 2020M1-2020M12

X13 Example: U.S. Unemployment Rate

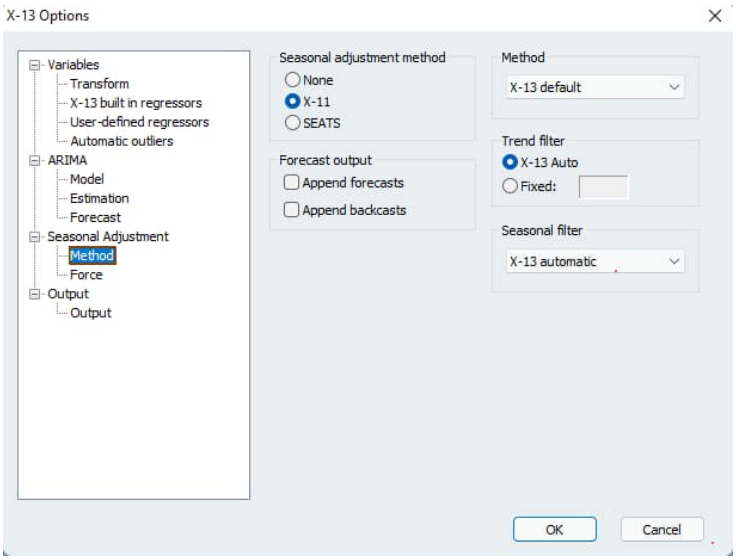
ARIMA Model Setting:



Select X-11 Auto as the ARIMA Method…Note that you can control the ARIMA models searched over…

X13 Example: U.S. Unemployment Rate

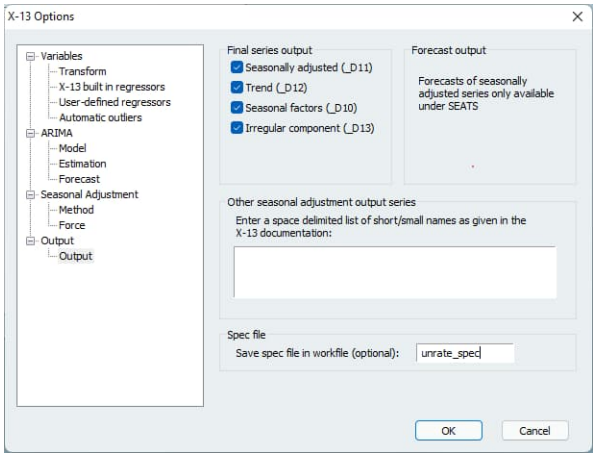
Seasonal Adjustment Setting:



Select X-11 Auto as the seasonal adjustment method.

X13 Example: U.S. Unemployment Rate

Output Setting:



Select the Output branch and check all four of the final series output boxes so that EViews will import the seasonally adjusted values, the trend values (D12), the seasonal factors (D10) and the irregular components (D13) into the workfile.

Also, the specification file that is sent to the X13 seasonally adjustment program will be written to an object called “unrate_spec” in the workfile. Click on this object to see the contents.

X13 Example: U.S. Unemployment Rate

After you click on OK, EViews will run the X-13ARIMA-SEATS program using the specified settings and will display the text output produced by the program in the series window.

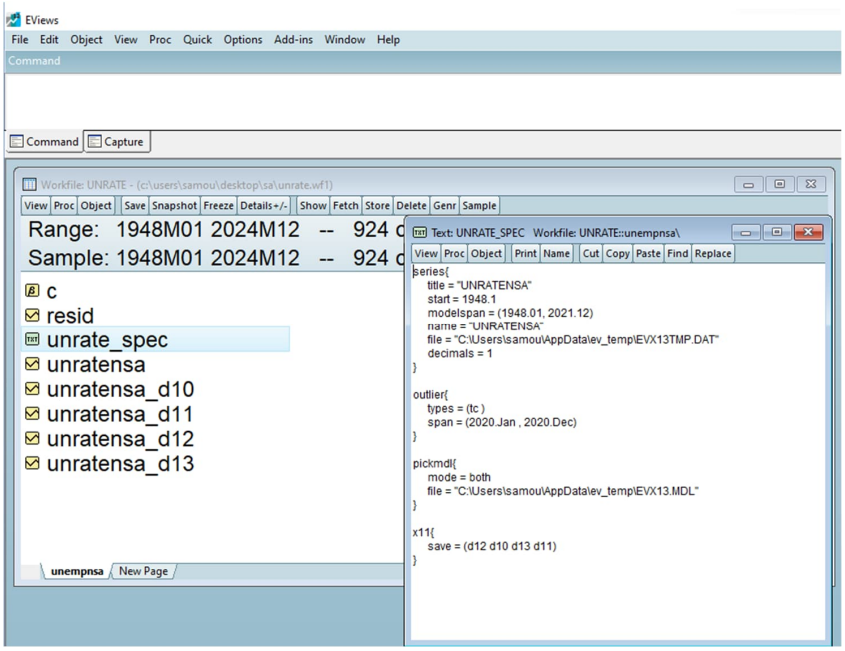
In addition, EViews will save four series UNRATENSA_D10, UNRATENSA_D11, UNRATENSA_D12 and UNRATENSA_D13 in the workfile (see slide 62).

The X13 specification file will be written to a text file object called “unrate_spec”. It contains the actual commands sent to the X13 census program.

UNRATENSA_D11 contains the final seasonally adjusted series.

Select this series along with the original UNRATENSA, right-click and Open/As group, then select View/Graph and click on OK to display a line graph of the adjusted and unadjusted series:

EViews Workfile After X13 Run



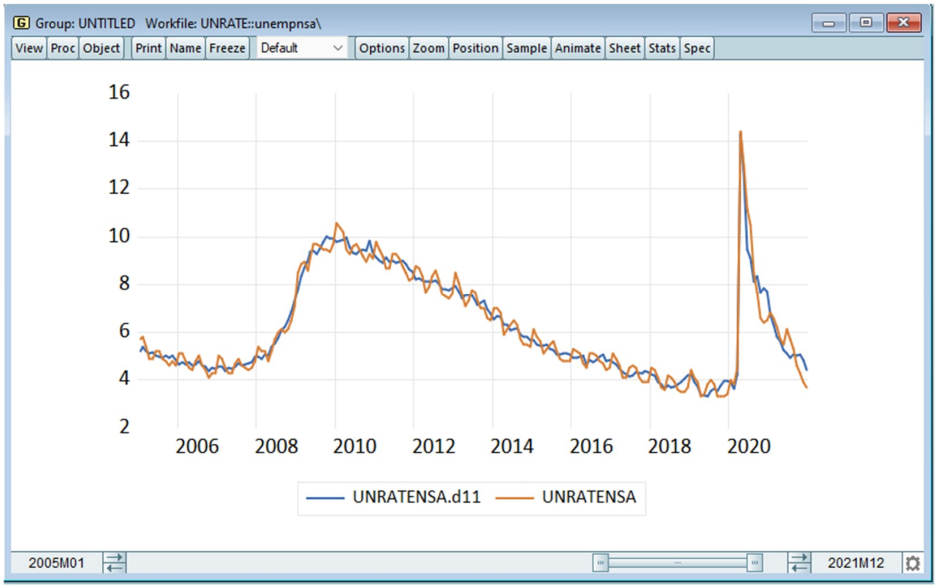
Actual versus Seasonally Adjusted UE, 2005-2021

Group: UNTITLED Workfile: UNRATE::unempnsa\
View Proc Object Print Name Freeze Default Sort Edit+/- Sm

	UNRATEN...	UNRATENSA
	UNRATEN...	UNRATENSA
2005M01	5.1984	5.7
2005M02	5.3842	5.8
2005M03	5.2080	5.4
2005M04	5.1163	4.9
2005M05	5.1425	4.9
2005M06	5.0043	5.2
2005M07	4.9879	5.2
2005M08	4.9343	4.9
2005M09	5.0071	4.8
2005M10	4.9207	4.6
2005M11	5.0126	4.8
2005M12	4.8120	4.6
2006M01	4.6563	5.1
2006M02	4.7313	5.1
2006M03	4.6461	4.8
2006M04	4.7173	4.5
2006M05	4.5973	4.4
2006M06	4.6359	4.8
2006M07	4.7816	5
2006M08	4.6239	4.6
2006M09	4.5765	4.4
2006M10	4.3835	4.1
2006M11	4.5120	4.3
2006M12	4.4743	4.3
2007M01		

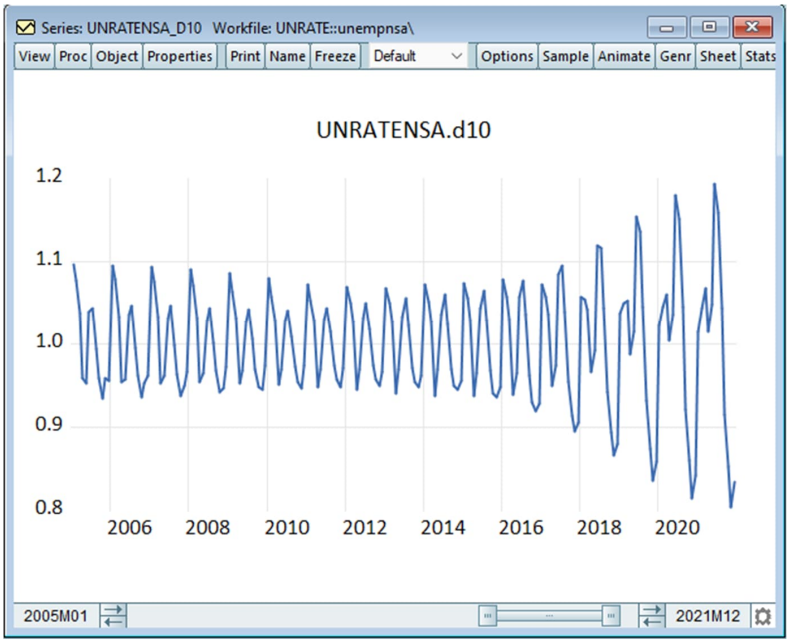
USA: Actual versus SA Unemployment, 2005-2021

The blue line, unratensa.d11, is the seasonally adjusted unemployment rate…

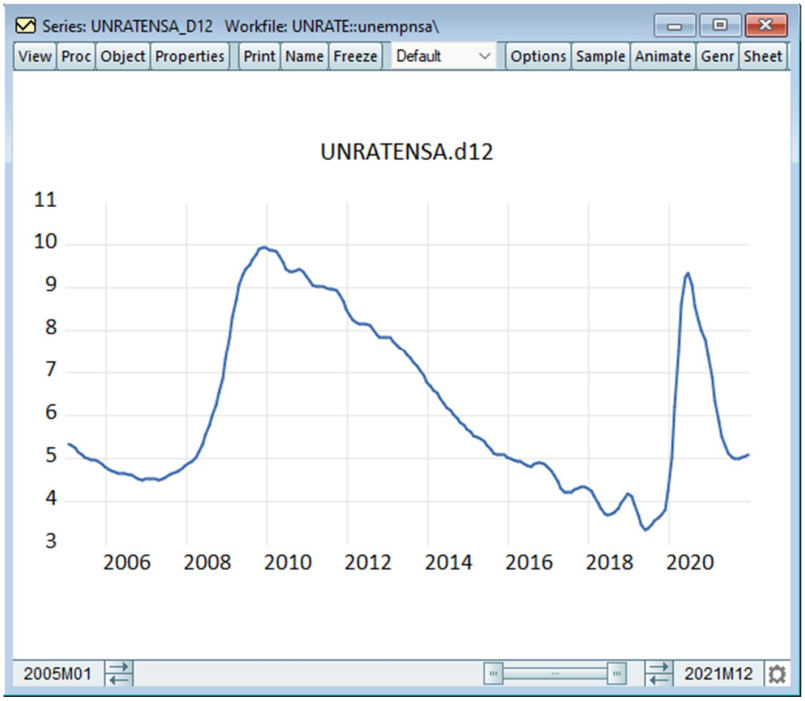


Seasonal Factors: D10

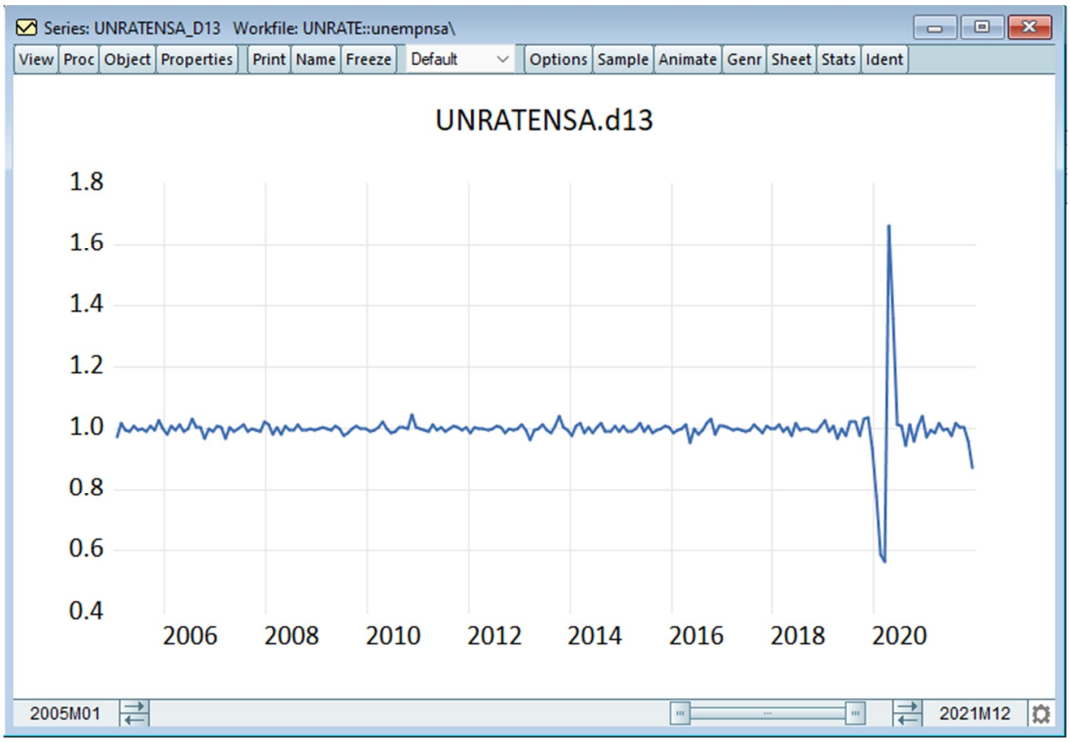
Note that the seasonal factors are expressed relative to 1.0 (for no-seasonal adjustment). This implies that the X-13 routine elected to use a multiplicative model.



Trend+Cycle Term: D12



Irregular Component: D13



References

IMF QNA Manual 2017 Edition_Chapter 7 (Seasonal Adjustment)

U.S. Census Bureau (2020), X-13ARIMA-SEATS Reference Manual, Statistical Research Division, U.S. Burea of the Census, Washington, DC.

EViews Help File (2020), Topic: Seasonal Adjustment.

Denton, F. (1971) “Adjustment of Monthly or Quarterly Series to Annual Totals: An Approach Based on Quadratic Minimization,” Journal of the American Statistical Association. 82, pp. 99–102

Cholette, Pierre A and Estela Bee Dagum (1994) “Benchmarking time series with autocorrelated survey errors”. In: International Statistical Review/Revue Internationale de Statistique, pp. 365– 377.